

Task 2. Engel Expansion

Available marks: 16

The Engel expansion of a positive real number x is the unique non-decreasing integer sequence $a_1, a_2, a_3 \dots$ such that

$$x = \frac{1}{a_1} + \frac{1}{a_1 a_2} + \frac{1}{a_1 a_2 a_3} + \dots$$

For example,

$$1.8 = \frac{1}{1} + \frac{1}{1 \times 2} + \frac{1}{1 \times 2 \times 2} + \frac{1}{1 \times 2 \times 2 \times 5} = 1 + 0.5 + 0.25 + 0.05$$

so the Engel expansion of 1.8 is the sequence 1 2 2 5

Write a program to display the Engel expansions of one or more real numbers, each of which has up to 16 significant digits. Care should be taken that rounding errors do not produce an incorrect sequence (such as 1 2 2 6 6 6 6 6 6 6 6 6 6 6... = 1.7999999999... for 1.8).

Input consists of one number per line, with the first line containing the number of numbers (maximum 20). Output should show each number, its Engel expansion, the calculated sum of reciprocals and the difference between the sum and the number to confirm the accuracy of the expansion. In some cases the difference may legitimately be tiny but non-zero.

Example input/output

<i>Input:</i> <pre>3 1.8 3.141592653589793 2.4</pre>	<i>Output:</i> <pre>1.8: 1 2 2 5; sum = 1.8, diff = 0.0000e+00 3.141592653589793: 1 1 1 8 8 17 19 300 1991 2499; sum = 3.1415926535897931, diff = 0.0000e+00 2.4: 1 1 3 5; sum = 2.4000000000000004, diff = -4.4409e-16</pre>
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Assessment

There is one test file `task2data.txt`. When you believe you have completed this task show the judges the program and its output when run with this data. Partial marks are awarded if most, but not all answers are correct.

Source: Wolfram Mathworld

mathworld.wolfram.com/EngelExpansion.html